

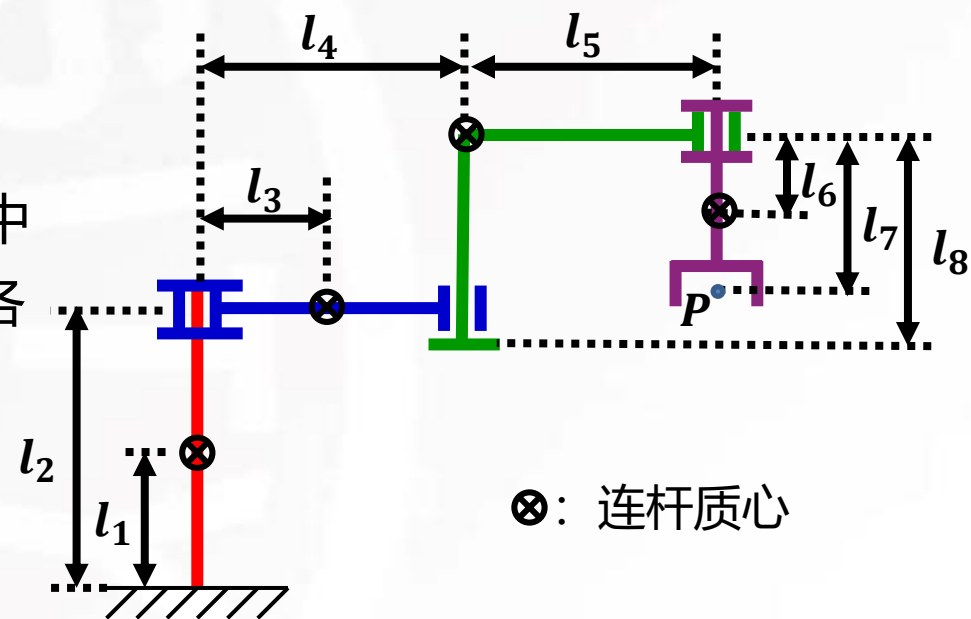
测试2

如图所示，一空间RPR机器人，各个结构尺寸分别为： $L_1 = 1$, $L_2 = 2$, $L_3 = 1$, $L_4 = 2$, $L_5 = 2$, $L_6 = 0.5$, $L_7 = 1$, $L_8 = 20$ ，红色杆件质量 $m_1 = 30$ ，蓝色杆件质量 $m_2 = 10$ ，绿色杆件质量 $m_3 = 10$ ，紫色杆件质量 $m_4 = 1$ ，各杆件质心处的转动惯量分别为：

$$I_{c1} = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 40 & 0 \\ 0 & 0 & 60 \end{bmatrix}, I_{c2} = \begin{bmatrix} 10 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 30 \end{bmatrix}, I_{c3} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 10 & 0 \\ 0 & 0 & 20 \end{bmatrix}, I_{c4} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

1. 画出该机器人坐标系，并给出此机器人的DH参数表 (15分)
2. 求取此机器人基础雅可比矩阵 (20分) 和工具雅可比矩阵 (10分)
3. 已知当此机器人构型为 $q = [0 \ 0 \ \pi/4]^T$ 时，机器人手抓中心 P 点处受外力 ${}^0F = [0 \ 0 \ 10]^T$ 时处于静力平衡，求取各个关节输出力/力矩。 (15分)
4. 求取此机器人标准形式的动力学方程 (40分)

$$M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \Gamma$$



测试2答案 (1)

1. 画出该机器人坐标系，并给出此机器人的DH参数表 (15分)

	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	$L_4 = 2$	d_2	0
3	0	$L_5 = 2$	0	θ_3
4	0	0	$-L_7 = -1$	0

2. 求取此机器人基础雅可比矩阵 (15分) 和工具雅可比矩阵 (10分)

$${}^0P_n = [(L_4 + L_5) \cos \theta_1 \quad (L_4 + L_5) \sin \theta_1 \quad d_2]^T$$

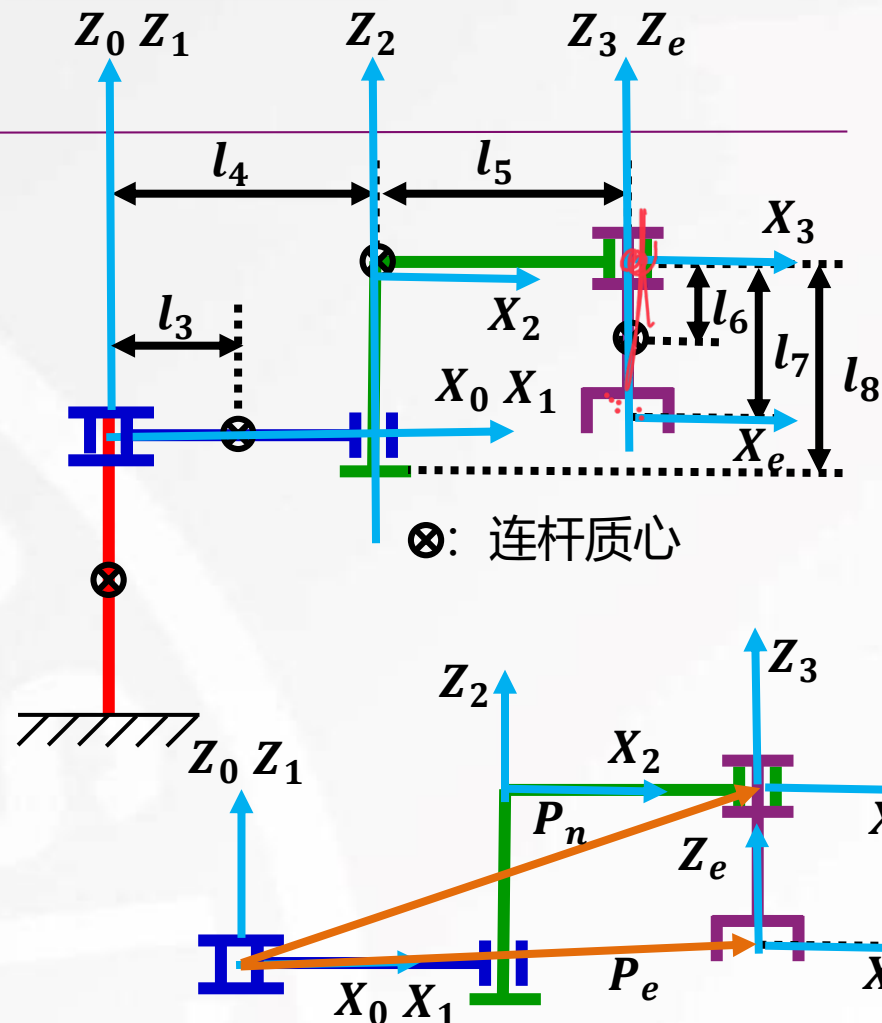
$${}^0P_e = [(L_4 + L_5) \cos \theta_1 \quad (L_4 + L_5) \sin \theta_1 \quad d_2 - L_7]^T$$

$$J_v = \frac{\partial {}^0P_n}{\partial q} = \begin{bmatrix} -(L_4 + L_5) \sin \theta_1 & 0 & 0 \\ (L_4 + L_5) \cos \theta_1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$${}^0Z_1 = {}^0Z_2 = {}^0Z_3 = {}^0Z_e = [0 \quad 0 \quad 1]^T$$

$$J_w = J_{ew} = [{}^0Z_1 \quad 0 \quad {}^0Z_3] = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$J_0 = J_e = \begin{bmatrix} -(L_4 + L_5) \sin \theta_1 & 0 & 0 \\ (L_4 + L_5) \cos \theta_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$



测试2答案 (2)

3. 已知当此机器人构型为 $q = [0 \ 0 \ \pi/4]^T$ 时, 机器人手抓中心受外力 ${}^0F = [0 \ 0 \ 10]^T$ 时处于静力平衡, 求取各个关节输出力/力矩。(15分)

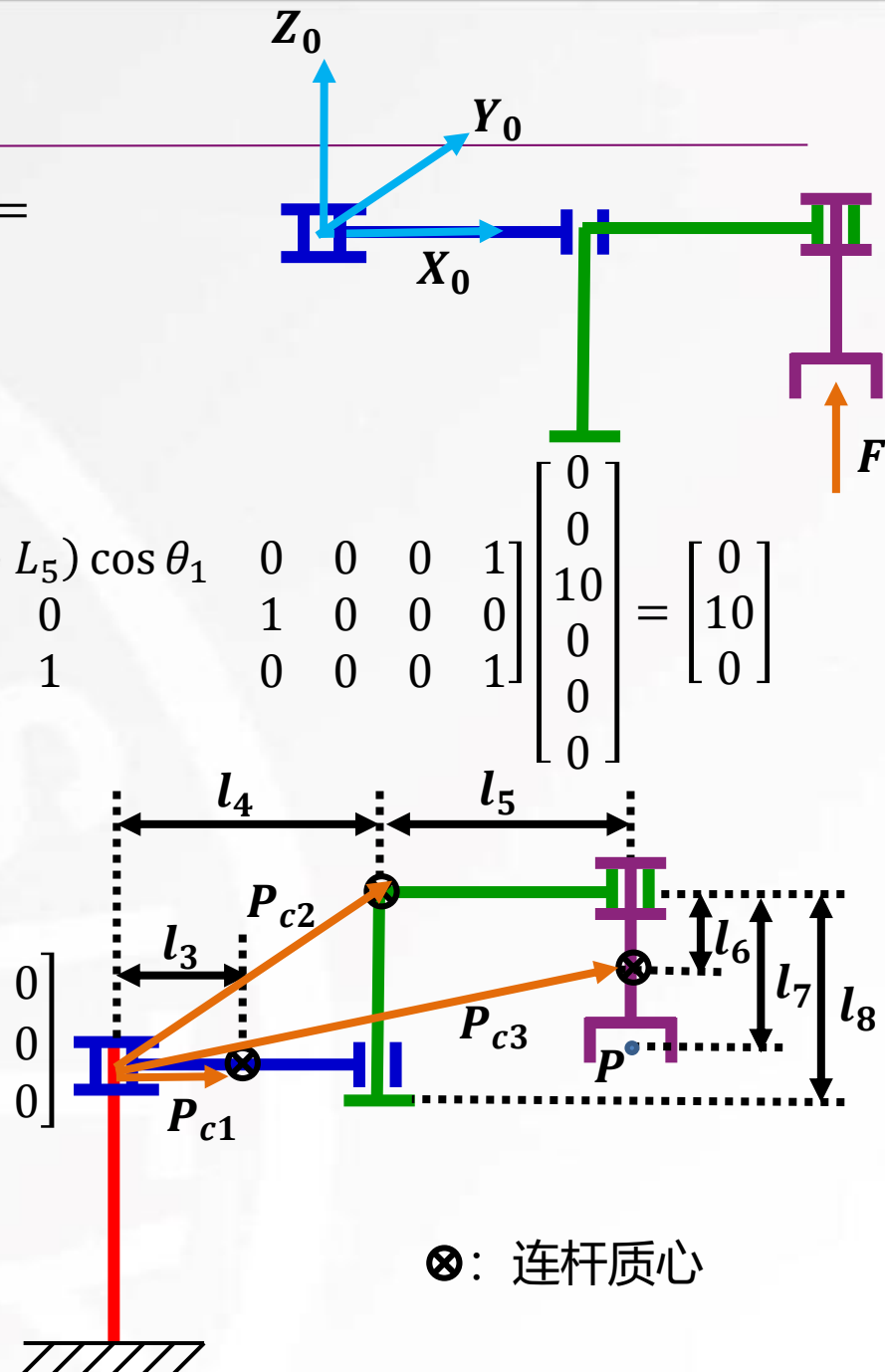
$$J_e = \begin{bmatrix} -(L_4 + L_5) \sin \theta_1 & 0 & 0 \\ (L_4 + L_5) \cos \theta_1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \Rightarrow \tau = J_e^T {}^0F = \begin{bmatrix} -(L_4 + L_5) \sin \theta_1 & (L_4 + L_5) \cos \theta_1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 10 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 10 \\ 0 \end{bmatrix}$$

4. 求取此机器人标准形式的动力学方程 (40分)

$${}^0P_{c1} = \begin{bmatrix} l_3 c1 \\ l_3 s1 \\ 0 \end{bmatrix}, {}^0P_{c2} = \begin{bmatrix} l_4 c1 \\ l_4 s1 \\ d_2 \end{bmatrix}, {}^0P_{c3} = \begin{bmatrix} (l_4 + l_5) c1 \\ (l_4 + l_5) s1 \\ d_2 - l_6 \end{bmatrix}$$

$${}^0J_{vc1} = \begin{bmatrix} -l_3 s1 & 0 & 0 \\ l_3 c1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, {}^0J_{vc2} = \begin{bmatrix} -l_4 s1 & 0 & 0 \\ l_4 c1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, {}^0J_{vc3} = \begin{bmatrix} -(l_4 + l_5) s1 & 0 & 0 \\ (l_4 + l_5) c1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$${}^0J_{wc1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, {}^0J_{wc2} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, {}^0J_{wc3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$



测试2答案 (3)

4. 求取此机器人标准形式的动力学方程 (40分)

$$M(q) = \sum_{i=1}^n J_{vci}^T m_i J_{vci} + J_{\omega ci}^T I_{ci} J_{\omega ci} = \begin{bmatrix} m_2 l_3^2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{c2}(3,3) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} m_3 l_4^2 & 0 & 0 \\ 0 & m_3 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{c3}(3,3) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ + \begin{bmatrix} m_4(l_4 + l_5) & 0 & 0 \\ 0 & m_4 & 0 \\ 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} I_{c3}(3,3) & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 109 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$b_{i,j,k} = \frac{\partial m_{i,j}}{\partial q_k} = 0, \quad c_{ijk} = \frac{1}{2}(b_{i,j,k} + b_{i,k,j} - b_{j,k,i}) = 0 \quad V(q, \dot{q}) = \begin{bmatrix} c_{111} & c_{122} & \cdots & c_{1nn} \\ c_{211} & c_{222} & \cdots & c_{2nn} \\ \vdots & \vdots & \ddots & \vdots \\ c_{n11} & c_{n22} & \cdots & c_{nnn} \end{bmatrix} \begin{bmatrix} \dot{q}_1^2 \\ \dot{q}_2^2 \\ \vdots \\ \dot{q}_n^2 \end{bmatrix} + \begin{bmatrix} 2c_{112} & 2c_{113} & \cdots & 2c_{1(n-1)n} \\ 2c_{212} & 2c_{213} & \cdots & 2c_{2(n-1)n} \\ \vdots & \vdots & \ddots & \vdots \\ 2c_{n12} & 2c_{n13} & \cdots & 2c_{n(n-1)n} \end{bmatrix} \begin{bmatrix} \dot{q}_1 \dot{q}_2 \\ \dot{q}_1 \dot{q}_3 \\ \vdots \\ \dot{q}_{n-1} \dot{q}_n \end{bmatrix} = 0$$

$$G(q) = -(m_1 J_{vc1}^T g + m_2 J_{vc2}^T g + \cdots + m_n J_{vcn}^T g) \\ = -m_2 \begin{bmatrix} -l_3 s1 & l_3 c1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} - m_3 \begin{bmatrix} -l_4 s1 & l_4 c1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} - m_4 \begin{bmatrix} -(l_4 + l_5) s1 & (l_4 + l_5) c1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} = (m_3 + m_4) g \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 109 & 0 & 0 \\ 0 & 11 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \\ \ddot{\theta}_3 \end{bmatrix} + 11g \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \tau_1 \\ F_2 \\ \tau_3 \end{bmatrix}$$